

Kryptos K4 via modified VIK CUT technique followed by Double Transposition via Tableaus 21 and 17.

Here is the description of the K4 "VIK CUT" and "ADD" to the front procedure as I believe described by the artist in the 7x14 backwards version of K4. It is immediately followed by the double transposition procedure which, among other things, demonstrates the linkage between the misspellings at positions 57 (L to Q) of K1, and 116 of K2 (O to U). This serves to link both LO to JFK and 57, possibly the year, to UU=U2. You can see the 7x14 table with the VIK CUT ADD instructions, the VIG O hint, the OSS arrow, and the aligned doublets in the file 7x14 in this directory. Here is how it works:

First, divide K4 into 16 blocks according to the message statistics series: 63 369 336 97, such that this equals 97 which is the length of K4 [$6+3+3+6+9+3+3+6+9+(7^2=7+7+7+7+7+7)$]:

1.OBKRUO (6) 2.XOG(3) 3.HUL(3) 4.BSOLIF(6) 5.BBWFLRVQQ(9) 6.PRN(3) 7.GKS(3) 8.SOTWTQ(6) 9. SJQSSEKZZ(9)= 48

10. WATJKLU(7) 11. DIAWINF(7) 12. BNYPVTT(7) 13.MZFPKWG(7) 14.DKZXTJC(7) 15.DIGKUH(7) 16.AUEKCAR (7)=49

The "VIK CUT" in K4 is constructed to get an "A to Z" match-up with the first columnar transposition so the blocks 16 to 10 are placed 16 to 1, 15 to 2 etc. until 10 to 7. The remaining blocks (original 1-9) are now 8 to 16 in their original order so the text order is:

AUEKCARDIGKUHDKZXTJCMZFPKWGBNYPVTTDIAWINWATJKLUOBKRUOXOGHULBSOLIFBBWFLRVQQP
RNGKSSOTWTQSJQSSEKZZ

This string is placed into a set of columnar transposition tables. The hints for lengths of the keywords were provided by the distances between the organized doublets in K4: 7 7 10 4 21 which also provides the hint for alignment of the 7x14 array discussed previously. Continue $7+10=17 + 4=21$. The keywords themselves were theorized based on a "direction finding" simulation exercise using the direction of the Kryptos compass (249 degrees as the needle points on the dial out of Langley/DC) and triangulated with the length, in degrees, of K4, at 97 West, (249 rhumbline from Langley at 97West is Dallas at 33N, 97W), and noting the length of K1 at 63, the placement of misspellings involving the letters L.O. , misspellings in columns 11 and 22, the "extra L" read over to O, the number and placement patterns of the question marks, and the shape of various items on the Sculptor's design model, among other things.

First transposition:

J	O	H	N	F	I	T	Z	G	E	R	A	L	D	K	E	N	N	E	D	Y
11	17	9	14	7	10	19	21	8	4	18	1	13	2	12	5	15	16	6	3	20
K	V	I	L	W	F	O	S	P	U	N	A	X	A	K	T	I	W	Z	G	S
L	Q	A	B	G	W	T	E	V	D	G	U	O	R	R	J	F	F	F	K	J
U	Q	W	S	B	A	W	K	T	K	K	E	G	D	U	C	B	L	P	U	Q
O	P	I	O	N	T	T	Z	T	Z	S	K	H	I	O	M	B	R	K	H	S
B	R	N	L	Y	J	Q	Z	D	X	S	C	U	ZZ	ZZ	ZZ	ZZ	ZZ	ZZ	ZZ	ZZ

Note the vertical **JKL** in the upper left hand corner of this chart. My theory is that this is a confirmation signal to follow this transposition (Tableau 21) with Tableau 17, keyword JOHNKENNEDYLANCER. Lancer was JFKs Secret Service Cryptonym. We take the results of the first transposition out by row and put it into the columns of Tableau 17 by column number. ZZ means a block box.

Second transposition:

J	O	H	N	K	E	N	N	E	D	Y	L	A	N	C	E	R
8	15	7	11	9	4	12	13	5	3	17	10	1	14	2	6	16
F	R	O	C	S	I	O	T	Q	N	Z	K	K	H	F	E	N
K	K	R	B	B	W	P	Z	A	A	D	K	V	I	O	V	L
J	H	R	L	A	Z	I	T	B	X	X	E	I	O	S	D	Y
U	S	J	P	W	G	O	Z	G	A	S	G	L	M	P	G	J
Q	B	F	U	K	S	N	S	W	K	C	D	W	B	U	U	Q
W	R	F	Q	T	L	T	K	T	T	U	U	ZZ	ZZ	ZZ	ZZ	ZZ

After completing Tableau 21 we note the curious fact that in the upper left hand corner it now reads vertically JFK, seemingly pointing back to Tableau 21? Note further that the text when taken out by rows starts FRO and ends TUU. JFK points to JKL which points to JFK and FRO and TUU (shouldn't it be to and fro?). Perhaps this is a little suspicious. Let's see if there is another way to validate the charts and this methodology as serving a designed function in Kryptos.

We will test the transposition algorithm on the first three letters written into Tableau 21 to see if this matches up with any other known design feature, like YAR, the misspellings, etc: The first three letters are AUE.

A is letter 1 written into Tableau 21 in column number 1. It becomes letter 12 when taken out by row and put into Tableau 17 as second letter in column 3, and is taken out at the end as number 27. U is taken out of Tableau 21 as number 33 and put into Tableau 17 at the bottom of column number 6. This causes it to end up as number 84, etc.

Into JFK (T21)	Out of JFK (T21)	Into JKL (T17)	Out of JKL (T17)	Position relative to Number 1
A=1	12	12 (column 3)	27	27-27=0
U=2 (U2)	33	33 (column 6)	84	84-27=57 57 is L to Q in K1
E=3	54	54 (column 10)	46 mod 97 (46+97=143)	143-27=116 116=O to U in K2

So the double transposition routine algorithm (Tableaus 21 and 17 in this exact keyword order) appears to have a validation design in the positioning of the misspellings? These also link L.O. to JFK and 57 to U2 in case that interests anyone. Likewise, the columns in which misspellings occur (11 22 etc.) contain the letter sequence KAIN inside the K4 segment with a few interesting interpretations applicable to both situations.

Note that the testing of the double transposition algorithm against the misspelling positions 57 and 116, on a mod97 basis has nothing to do with the modified VIK CUT procedure described above. Other tests show that this is the procedure that fits the "A to Z" theory and the JKL and JFK "signals" in the upper left hand corner, not to mention the text FRO and TUU.

The VIK CUT and double transposition routine described above has been placed for discovery into one other "double art work" routine made by Jim Sanborn, in effect providing a link between Kryptos and the Cyrillic Projector. How this works is described in another note entitled "MEDUSA and FORKYS etc" in Kent R Files. In this case drawing attention to the Quagmire 2 decryption method used in the Cyrillic Projector which was solved by Frank Corr, an official of the American Cryptogram Association.

It is not presently known whether this transposition routine has a role in the solution of the remaining unsolved 97 letters, is a dead-end, or side track design of some sort, or as was recently suggested at the NSA Symposium, perhaps is part of telling a Kryptos and/or CIA Story via Classic Cryptography tools, or a combination of the above./KR